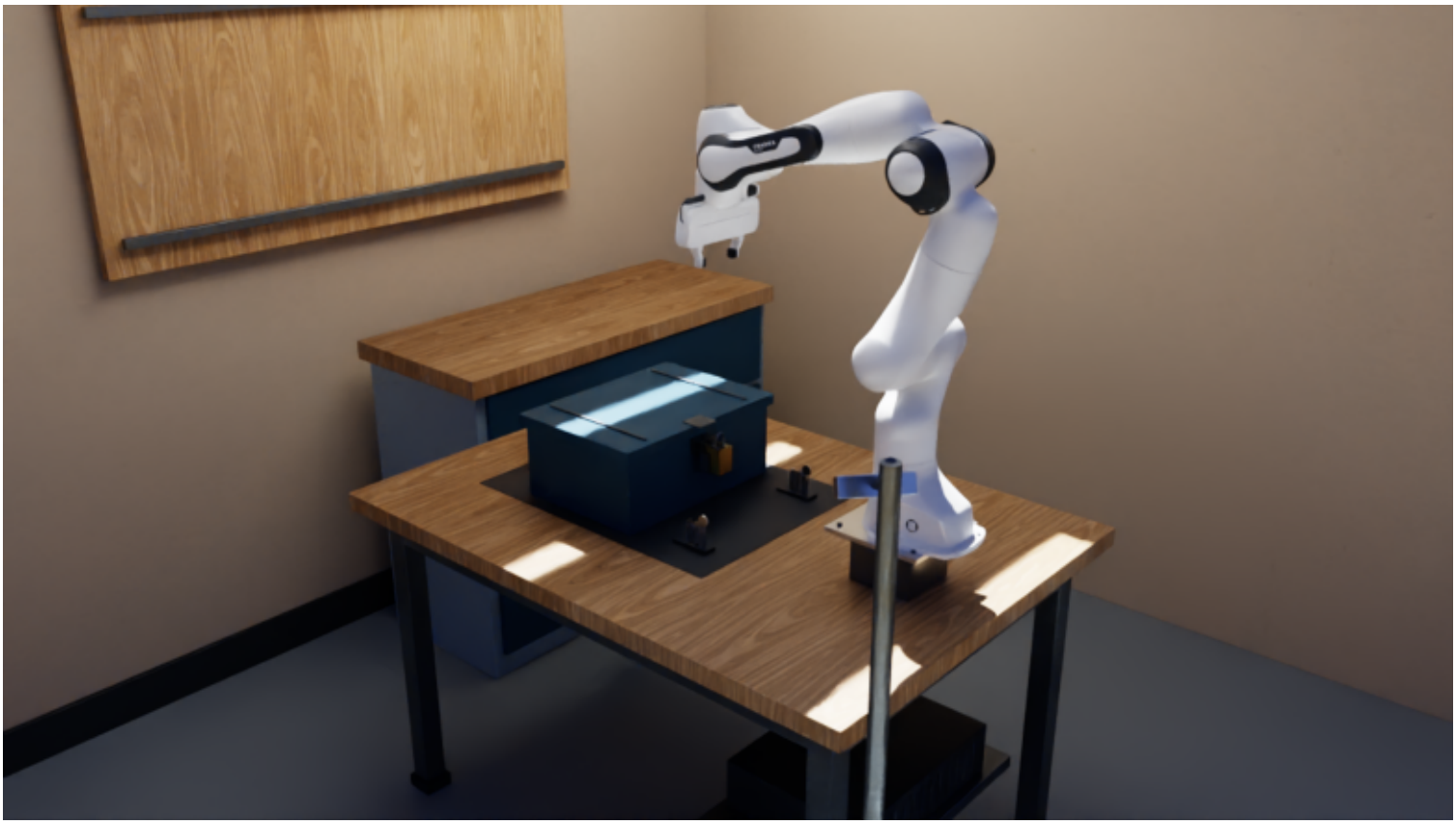
A workshop lock-and-key task

Standard Panda, contact-based pin mechanics, authenticated remote operation



One Panda handles the correct key, inserts and turns it, then grips and raises the shackle. The second key must remain undisturbed. The original task does not require opening the box lid. This report covers creating and qualifying the environment; it does not claim an autonomous robot solution.

Engine / batch	Newton 1.5.0 / 1 environment
Physics / recording	1920 Hz physics + history / 30 Hz RGB
Full-runtime cost	99.37 wall s per simulated s
Evidence	Key discrimination, shackle release, robot grip/lift/release

The ideal guides and reduced shackle latch are disclosed; this is not a security-certified replica of a commercial padlock.

Scene, objects and geometry

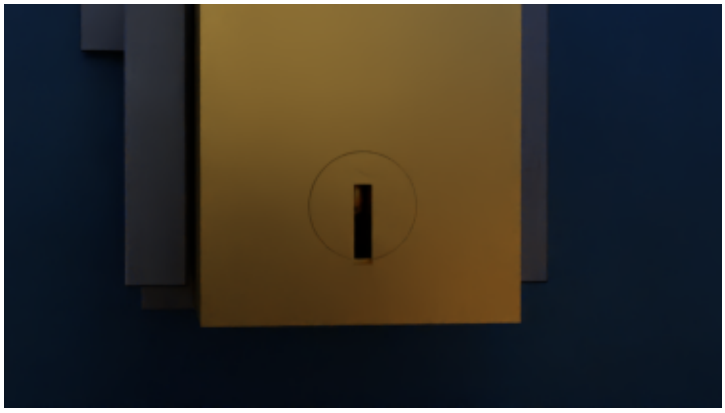
Visible supports and colliders; manufacturer robot rather than invented kinematics



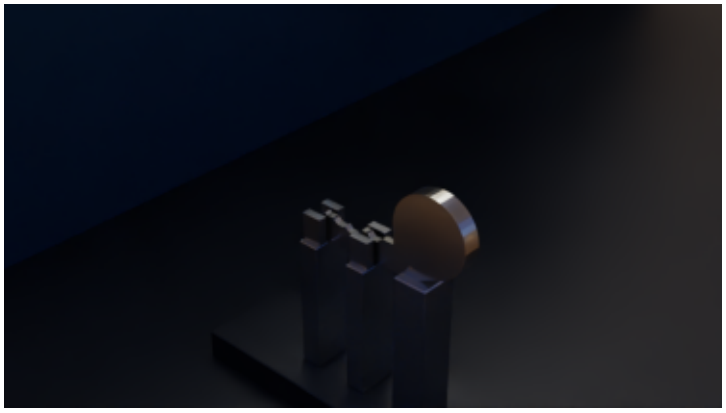
Lock on the hasp and staple of the box



Newton workshop



Open keyway in the actual casing collider



Solid key bow and physical cradles

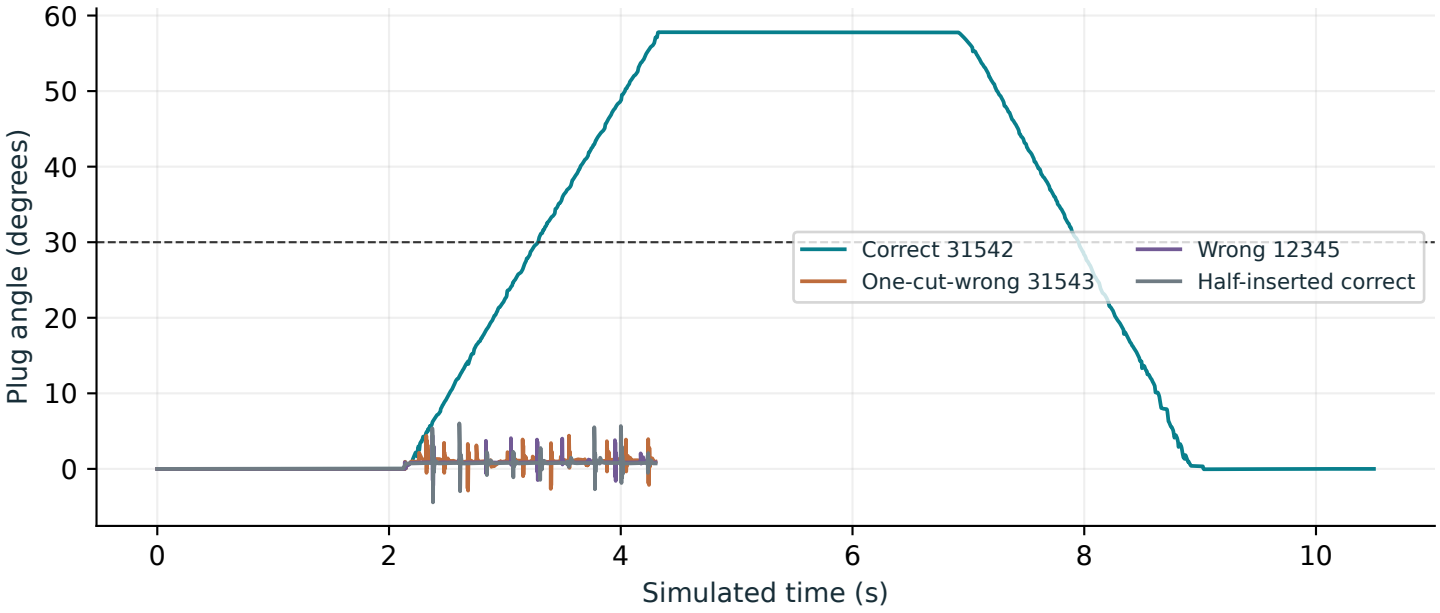
750 mm workbench, bolted robot riser, fixed lock bracket, steel/painted box and physically represented external camera. Task/robot relative starting poses follow the original task layout.

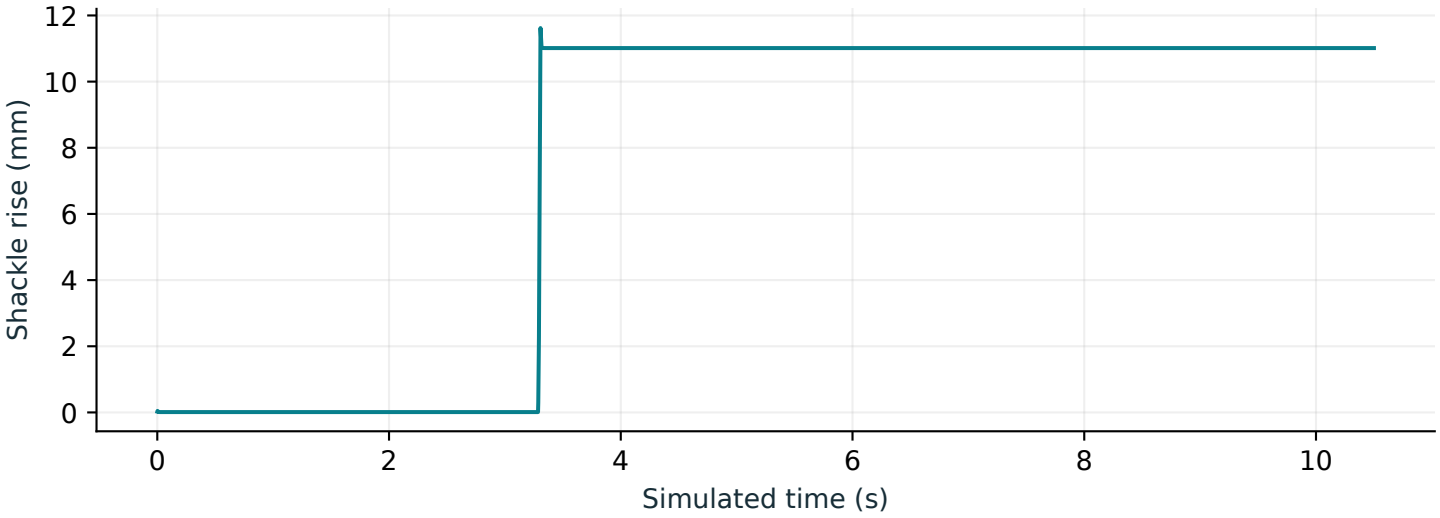
The five-pin plug is 12.7 mm diameter and 28 mm long. Round bores give the plug complete collision walls, so a one-cut-wrong key cannot rotate through a gap. Cradle surfaces follow the actual solid bow underside. The visible casing has a real keyway.

The staple and lid do not overlap initially. The rigid hasp does not articulate off the staple: this is not a fully operable chest, and lid opening is not the benchmark goal.

Key discrimination and release

Full assembled scene; a bounded mechanical hand is explicit assistance



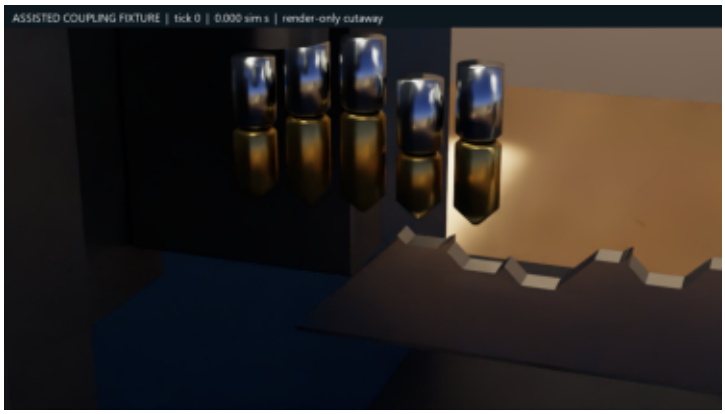


With 5 N upward force while locked, maximum shackle rise is 0.0106 mm. After correct-key rotation the shackle rises 11.62 mm. Maximum wrong-key angles: adjacent 4.40, supplied wrong 4.44, partial 6.02 degrees; none reaches the 30-degree release threshold.

Key assistance is a cylindrical guide with 26 N translation / 0.3 Nm torque caps. No key-code comparison authorizes rotation; key/pin/plug contact carries load. No pin-floor latching or collider disable is used. The shackle stop releases at measured angle; no contact-resolved cam is claimed.

Inside the physical mechanism

Render-only cutaway of the same continuous full-assembly cycle



Tick 0: key in the assisted insertion guide



Tick 6720: key pins have rotated with the plug

Five upper drivers translate along fixed-world bore axes. Five lower key pins translate in guides carried by the rotating plug. Their contact with the key cuts determines whether the shear line can clear. The cutaway hides the casing, shell and plug surface ONLY in the renderer; no collider is removed during the physical run.

Spring load acts on each driver through a native prismatic-joint actuator. At rest its target preload is approximately 0.3 N. Ordinary key torque is transmitted through collision contacts. The key's bitting string constructs geometry; it is not consulted by a rule that grants rotation.

Parameter	Implemented value / provenance
Pin diameter / adjacent cut step	2.921 / 0.5842 mm; inherited KW1 layout
Driver length	4.572 mm; LAB 0.180-inch option
Bottom pin lengths	4.3688 to 7.8740 mm; LAB depths 1..7
Driver spring / damping	40 N/m / 0.6 Ns/m; inherited estimates
Bore / plug radial clearance	0.100 / 0.050 mm; authored model values

Ideal guide constraints stand in for bearing reactions. The shackle release is still a reduced angle-triggered latch, not a simulated locking dog/cam. Published values and inherited estimates are kept distinct; none of these values is presented as a new physical measurement.

Robot grip, lift and release

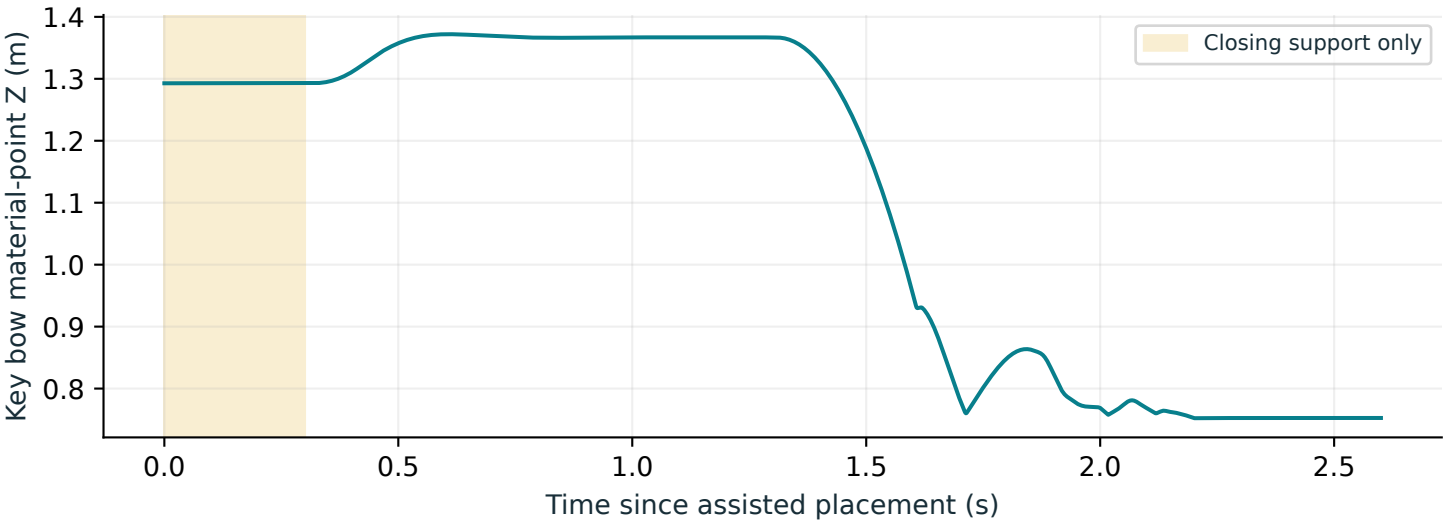
One continuous recorded fixture; no object-to-gripper weld



After closing on the key bow



After upward robot motion

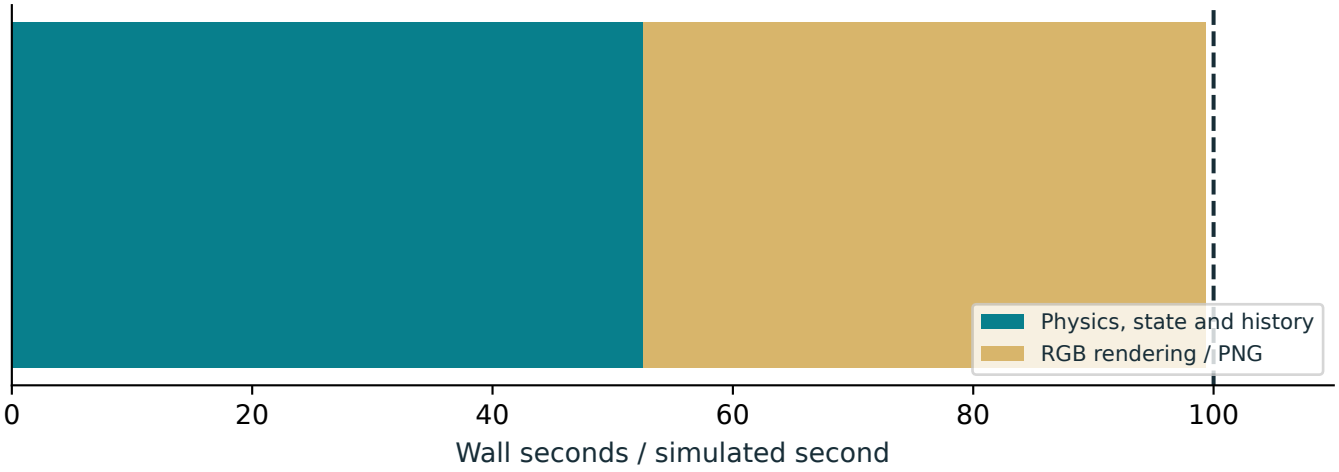


The key is placed between open fingers. Weight is supported only during the first 0.3 s of closing; support ends before lifting. An ordinary 100 mm upward IK command raises the tracked bow point by 74.08 mm. Opening releases the independent key onto the bench.

The mechanism and robot use different native contact-response settings to accommodate their scale. The 45 N per-finger controller cap is not presented as a measured Franka grip force. This is a contact/manipulation component check, not autonomous table pickup or a solved unlocking task.

Runtime and remote integration

Measured complete inventory, actual RGB and every-step privileged history



Batch 1, NVIDIA GeForce RTX 3090; 24 bodies, 236 shapes; 1920 ticks / 1.00 simulated seconds. 31 RGB frames at 1280 x 720 and 30 Hz. Full-inventory hold wall/sim 99.37, below the 100 target. Startup is separate: 27.04 s. Post-run archives, diagnostic stills and video export are also separate.

GAR create/state/obs/act/reset and an actual ephemeral HTTP server were exercised. Unauthorized requests fail; restricted keys cannot retrieve object state, geometry or privileged histories. Duplicate idempotency keys execute only once. Invalid actions do not advance ticks. Reset is explicit, IK delivers physical setpoints, and stale state cannot be paired with a new image.

Interface	Teacher	Restricted policy
Mounted RGB / joint encoders / robot actions	Yes	Yes
Exact object state / meshes / contact forces	Yes	No
Reward / per-step action histories	Yes	No
Checkpoint / variation catalog	Not implemented	Not implemented

History has bounded retention and explicit completeness. A process being alive is not evidence that physics is healthy. Faults stop movement; owner recovery does not replay an action or hide a reset.

Research, limitations and reproduction

Digital-only evidence; no physical calibration or autonomous-solver requirement

Pin lengths follow LAB/Kwikset service dimensions; C36000 specific gravity supports the 8,500 kg/m3 brass estimate. Pin preload, damping and lubricated friction remain disclosed inherited estimates where manufacturer data does not give exact values. The Panda comes from the pinned manufacturer description.

Limitations: rigid solids, ideal pin/bearing guides, reduced angle-triggered shackle latch, fixed lock and box, solid simplified key bow, non-articulating hasp, ideal external pinhole camera. No deformation, wear/fracture, lubrication film, exact product certification or collision-free IK guarantee is claimed.

Failed candidates include an isolated bare-blade return/withdrawal failure. A completed process is never itself called a physical pass. Videos identify every recorded tick and do not splice restored checkpoints.

Reproduce with the release: install it with the Isaac Sim 5.1 renderer, then start the task with the command below. Isaac renders the scene in its own interpreter; Newton 1.5.0 performs dynamics.

```
gar run lockkey --port 8080
```

Newton 1.5 collision and native solver documentation

LAB: original Kwikset replacement pins

Copper Development Association: C36000

Franka Hand product manual